

培道中學校

姓名：

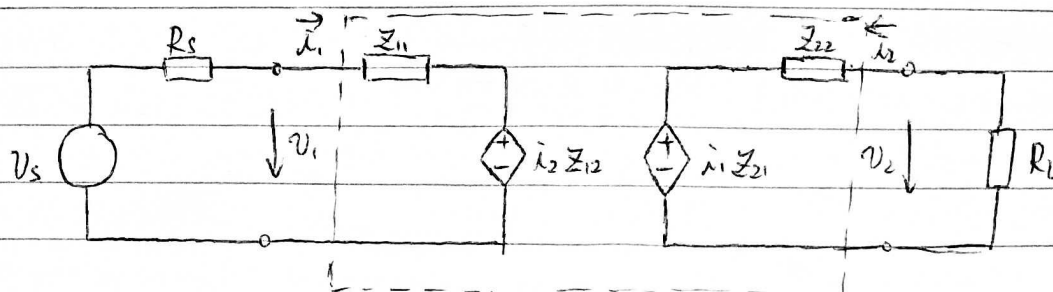
(號)

成績

年級：

科目：

作業 1.

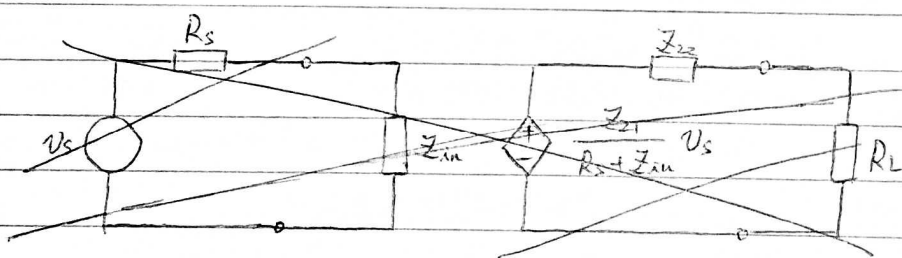


$$V_1 = i_1 Z_{11} + i_2 Z_{12}$$

$$= -i_1 Z_{11} + (-i_1 Z_{21} - i_2)$$

$$= i_1 Z_{11} + \left(\frac{-i_1 Z_{21}}{Z_{22} + R_L} \right) Z_{12} = i_1 \left(Z_{11} - \frac{Z_{12} Z_{21}}{Z_{22} + R_L} \right)$$

$$\Rightarrow Z_{in} = Z_{11} - \frac{Z_{12} Z_{21}}{Z_{22} + R_L}$$

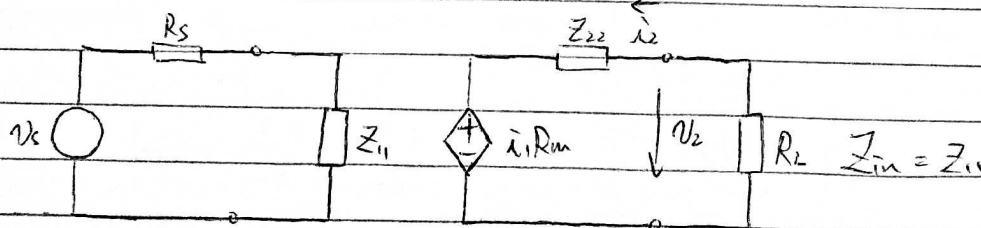


$$V_2 = i_2 Z_{22} + i_1 Z_{21}$$

$$= i_2 Z_{22} + \left(\frac{V_s - i_2 Z_{12}}{R_s + Z_{11}} \right) Z_{21} = \left(Z_{22} - \frac{Z_{12} Z_{21}}{R_s + Z_{11}} \right) i_2 + \frac{Z_{21}}{Z_{11} + R_s} V_s$$

$$\Rightarrow Z_{out} = Z_{22} - \frac{Z_{12} Z_{21}}{Z_{11} + R_s}, \quad V_{TH} = \frac{Z_{21}}{Z_{11} + R_s} V_s$$

考慮單向網絡 $Z_{12} = 0$, $Z_{21} = R_m$



$$\text{顯然 } Z_{out} = Z_{22}, \quad i_1 = \frac{V_s}{R_s + Z_{11}}, \quad V_{TH} = \frac{R_m}{R_s + Z_{11}} V_s, \quad V_2 = \frac{R_L}{Z_{22} + R_L} V_{TH} = \frac{R_L}{Z_{22} + R_L} \frac{R_m}{R_s + Z_{11}} V_s$$

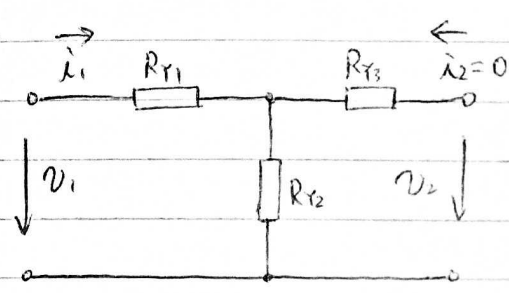
$$V_2 = i_2 Z_{22} + \frac{R_m}{R_s + Z_{11}} V_s$$

觀察 $V_2 = i_2 Z_{22} + \frac{R_m}{R_s + Z_{11}} V_s$

結合: $V_2 + i_2 R_L = 0$ 後, 可得 $i_2 = - \frac{1}{R_L + Z_{22}} \cdot \frac{R_m}{R_s + Z_{11}} V_s$

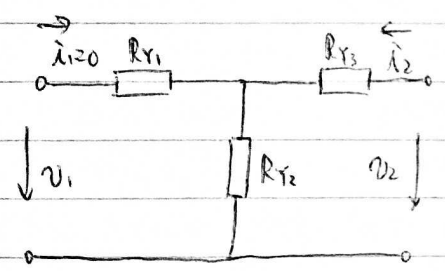
輸出電壓 經過回路 電阻所產生電流
 負載電阻 並向輸出電壓
 輸入回路 電流

作業 2.



此時有 $Z_{11} = \frac{V_1}{i_1} \Big|_{i_2=0} = R_1 + R_2$

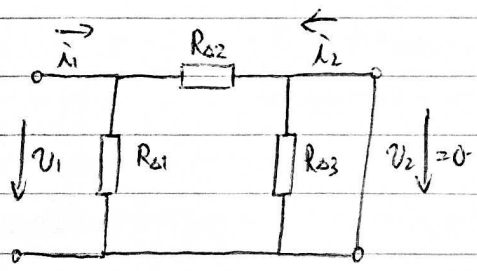
$Z_{21} = \frac{V_2}{i_1} \Big|_{i_2=0} = R_2$



此時有, $Z_{12} = \frac{V_1}{i_2} \Big|_{i_1=0} = R_2$

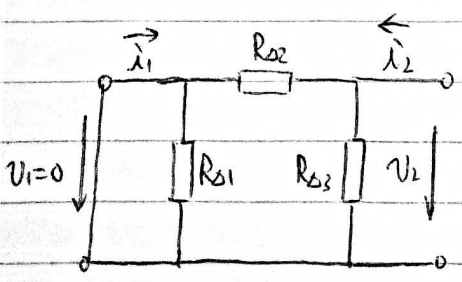
$Z_{22} = \frac{V_2}{i_2} \Big|_{i_1=0} = R_2 + R_3$

因此 $Z_F = \begin{pmatrix} R_1 + R_2 & R_2 \\ R_2 & R_2 + R_3 \end{pmatrix}$, $Y_F = Z_F^{-1} = \frac{1}{(R_1 + R_2)(R_2 + R_3) - R_2^2} \begin{pmatrix} R_2 + R_3 & -R_2 \\ -R_2 & R_1 + R_2 \end{pmatrix}$



此時有 $y_{11} = \frac{i_1}{V_1} \Big|_{V_2=0} = \frac{1}{R_1} + \frac{1}{R_2}$

$y_{21} = \frac{i_2}{V_1} \Big|_{V_2=0} = -\frac{1}{R_2}$



此時有 $y_{12} = \frac{i_1}{V_2} \Big|_{V_1=0} = -\frac{1}{R_2}$

$y_{22} = \frac{i_2}{V_2} \Big|_{V_1=0} = \frac{1}{R_2} + \frac{1}{R_3}$

年級：

科目：

1 因為

$$\begin{pmatrix} \frac{1}{R_{\Delta 1}} + \frac{1}{R_{\Delta 2}} & -\frac{1}{R_{\Delta 2}} \\ -\frac{1}{R_{\Delta 2}} & \frac{1}{R_{\Delta 2}} + \frac{1}{R_{\Delta 3}} \end{pmatrix} = \frac{1}{(R_{Y1}+R_{Y2})(R_{Y2}+R_{Y3})-R_{Y2}^2} \begin{pmatrix} R_{Y2}+R_{Y3} & -R_{Y2} \\ -R_{Y2} & R_{Y1}+R_{Y2} \end{pmatrix}$$

$$\Rightarrow \frac{1}{R_{\Delta 2}} = \frac{R_{Y2}}{(R_{Y1}+R_{Y2})(R_{Y2}+R_{Y3})-R_{Y2}^2} \Leftrightarrow R_{\Delta 2} = \frac{R_{Y1}R_{Y3}}{R_{Y2}} + R_{Y1}+R_{Y3}$$

又知把 Y_{Δ} 的第一、二行相減與 Y_Y 的第一、二行相加的結果一致

$$\text{所以 } \begin{pmatrix} \frac{1}{R_{\Delta 1}} & \frac{1}{R_{\Delta 3}} \end{pmatrix} = \frac{1}{R_{Y1}R_{Y3}+R_{Y2}(R_{Y1}+R_{Y3})} \begin{pmatrix} R_{Y3} & R_{Y1} \end{pmatrix}$$

$$\Leftrightarrow (R_{\Delta 1} \ R_{\Delta 3}) = \left(\frac{R_{Y1}R_{Y2}}{R_{Y3}} + R_{Y1}+R_{Y2} \quad \frac{R_{Y2}R_{Y3}}{R_{Y1}} + R_{Y2}+R_{Y3} \right)$$

$$\text{因此 } \begin{pmatrix} R_{\Delta 1} \\ R_{\Delta 2} \\ R_{\Delta 3} \end{pmatrix} = \begin{pmatrix} \frac{R_{Y1}R_{Y2}}{R_{Y3}} + R_{Y1}+R_{Y2} \\ \frac{R_{Y2}R_{Y1}}{R_{Y3}} + R_{Y3}+R_{Y1} \\ \frac{R_{Y1}R_{Y2}}{R_{Y3}} + R_{Y1}+R_{Y2} \end{pmatrix}$$

反之

$$Z_{\Delta} = Y_{\Delta}^{-1} = \frac{1}{(G_{\Delta 1}+G_{\Delta 2})(G_{\Delta 2}+G_{\Delta 3})-G_{\Delta 2}^2} \begin{pmatrix} G_{\Delta 2}+G_{\Delta 3} & -G_{\Delta 2} \\ -G_{\Delta 2} & G_{\Delta 1}+G_{\Delta 2} \end{pmatrix}$$

同時

$$Z_{\Delta} = Z_Y = \begin{pmatrix} R_{Y1}+R_{Y2} & -R_{Y2} \\ -R_{Y2} & R_{Y2}+R_{Y3} \end{pmatrix}$$

$$\text{所以 } R_{Y2} = \frac{G_{\Delta 2}}{G_{\Delta 1}G_{\Delta 2}+G_{\Delta 2}G_{\Delta 3}+G_{\Delta 3}G_{\Delta 1}} = \frac{R_{\Delta 2}R_{\Delta 1}}{R_{\Delta 1}+R_{\Delta 2}+R_{\Delta 3}}$$

$$\Rightarrow (R_{Y1}-R_{Y3}) = \frac{1}{G_{\Delta 1}G_{\Delta 2}+G_{\Delta 2}G_{\Delta 3}+G_{\Delta 3}G_{\Delta 1}} (G_{\Delta 23}-G_{\Delta 01})$$

$$\Leftrightarrow (R_{Y1} \ R_{Y3}) = \begin{pmatrix} R_{\Delta 1}R_{\Delta 2} & R_{\Delta 2}R_{\Delta 3} \\ R_{\Delta 1}+R_{\Delta 2}+R_{\Delta 3} & R_{\Delta 1}+R_{\Delta 2}+R_{\Delta 3} \end{pmatrix}$$

若利用 ABCD 參量, 則對於 Y 形網絡

$$\begin{pmatrix} A_Y & B_Y \\ C_Y & D_Y \end{pmatrix} = \begin{pmatrix} 1 & R_{Y1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ G_{Y2} & 1 \end{pmatrix} \begin{pmatrix} 1 & R_{Y3} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & R_{Y1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & R_{Y3} \\ G_{Y2} & G_{Y2}R_{Y3}+1 \end{pmatrix} \\ = \begin{pmatrix} G_{Y2}R_{Y1}+1 & G_{Y2}R_{Y3}R_{Y1}+R_{Y3}+R_{Y1} \\ G_{Y2} & G_{Y2}R_{Y3}+1 \end{pmatrix}$$

對於 Δ 形網絡

$$\begin{pmatrix} A_\Delta & B_\Delta \\ C_\Delta & D_\Delta \end{pmatrix} = \begin{pmatrix} 1 & \\ G_{\Delta1} & 1 \end{pmatrix} \begin{pmatrix} 1 & R_{\Delta2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ G_{\Delta3} & 1 \end{pmatrix} = \begin{pmatrix} 1 & \\ G_{\Delta1} & 1 \end{pmatrix} \begin{pmatrix} G_{\Delta3}R_{\Delta2}+1 & R_{\Delta2} \\ G_{\Delta3} & 1 \end{pmatrix} = \begin{pmatrix} G_{\Delta3}R_{\Delta2}+1 & R_{\Delta2} \\ G_{\Delta1}G_{\Delta3}R_{\Delta2}+G_{\Delta3}+G_{\Delta1} & G_{\Delta1}R_{\Delta2}+1 \end{pmatrix}$$

$$\Rightarrow G_{Y2} = G_{\Delta1}G_{\Delta3}R_{\Delta2} + G_{\Delta3} + G_{\Delta1} = \frac{R_{\Delta1}R_{\Delta2}R_{\Delta3}}{R_{\Delta1}R_{\Delta3}}$$

$$\Leftrightarrow R_{Y2} = \frac{R_{\Delta1}R_{\Delta3}}{R_{\Delta1}+R_{\Delta2}+R_{\Delta3}} \quad \text{同法} \quad R_{\Delta2} = G_{Y2}R_{Y3}R_{Y1} + R_{Y3} + R_{Y1} = \frac{R_{Y3}R_{Y1}}{R_{Y2}} + R_{Y3} + R_{Y1}$$

$$G_{Y2}R_{Y1}+1 = G_{\Delta3}R_{\Delta2}+1 \Leftrightarrow R_{Y1} = \frac{G_{\Delta3}R_{\Delta2}}{G_{Y2}} = \frac{R_{\Delta2}}{R_{\Delta3}} \cdot \frac{R_{\Delta1}R_{\Delta3}}{R_{\Delta1}+R_{\Delta2}+R_{\Delta3}} = \frac{R_{\Delta1}R_{\Delta2}}{R_{\Delta1}+R_{\Delta2}+R_{\Delta3}}$$

$$G_{\Delta1}R_{\Delta2}+1 = G_{Y2}R_{Y3}+1 \Leftrightarrow G_{\Delta1} = \frac{G_{Y2}R_{Y3}}{R_{\Delta2}} = \frac{R_{Y3}}{R_{\Delta2}} \cdot \frac{R_{\Delta1}R_{\Delta2}}{R_{\Delta1}+R_{\Delta2}+R_{\Delta3}} = \frac{R_{\Delta1}R_{Y3}}{R_{\Delta1}+R_{\Delta2}+R_{\Delta3}}$$

$$\Leftrightarrow R_{\Delta1} = \frac{R_{\Delta2}}{G_{Y2}R_{Y3}} = \frac{R_{\Delta2}}{R_{Y3}} \left(\frac{R_{Y3}R_{Y1}}{R_{Y2}} + R_{Y3} + R_{Y1} \right) = \frac{R_{Y1}R_{Y2}}{R_{Y3}} + R_{Y1} + R_{Y2}$$

$$R_{Y3} = \frac{G_{\Delta1}R_{\Delta2}}{G_{Y2}} = \frac{R_{\Delta1}R_{\Delta3}}{R_{\Delta1}+R_{\Delta2}+R_{\Delta3}} \cdot \frac{R_{\Delta2}}{R_{\Delta1}} = \frac{R_{\Delta2}R_{\Delta3}}{R_{\Delta1}+R_{\Delta2}+R_{\Delta3}}$$

$$G_{\Delta3} = \frac{G_{Y2}R_{Y1}}{R_{\Delta2}} \Leftrightarrow R_{\Delta3} = \frac{R_{\Delta2}}{G_{Y2}R_{Y1}} = \frac{R_{Y2}}{R_{Y1}} \left(\frac{R_{Y3}R_{Y1}}{R_{Y2}} + R_{Y3} + R_{Y1} \right) = \frac{R_{Y2}R_{Y3}}{R_{Y1}} + R_{Y2} + R_{Y3}$$

姓名：

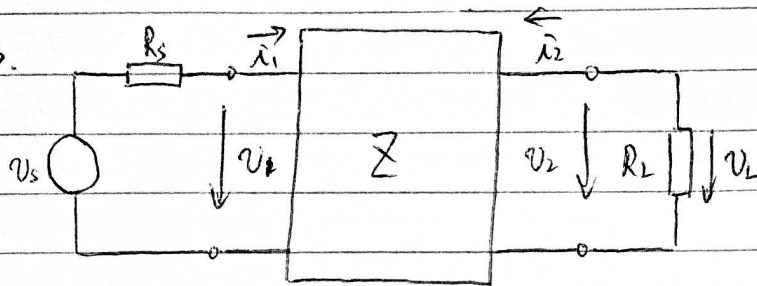
(號)

成績

年級：

科目：

作業3.



$$\begin{pmatrix} v_s - i_1 R_s \\ v_L \end{pmatrix} = \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ -\frac{v_L}{R_L} \end{pmatrix}$$

$$\Rightarrow v_L = i_1 z_{21} + \left(-\frac{v_L}{R_L}\right) z_{22} \Leftrightarrow i_1 = \frac{v_L + \frac{v_L}{R_L} z_{22}}{z_{21}} = \frac{R_L + z_{22}}{R_L z_{21}} v_L$$

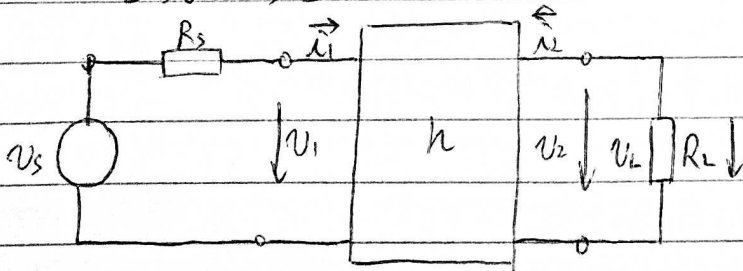
$$v_s - i_1 R_s = i_1 z_{11} + \left(-\frac{v_L}{R_L}\right) z_{12} \Leftrightarrow v_s = (z_{11} + R_s) \cdot \frac{R_L + z_{22}}{R_L z_{21}} v_L + \left(-\frac{v_L}{R_L}\right) z_{12}$$

$$\Rightarrow \frac{v_s}{v_L} = \frac{(z_{11} + R_s)(R_L + z_{22})}{R_L z_{21}} + \left(-\frac{z_{12}}{R_L}\right)$$

$$\Leftrightarrow H_{\text{雙向網絡}} = \frac{v_L}{v_s} = \frac{R_L z_{21}}{z_{12} z_{21} + (z_{11} + R_s)(R_L + z_{22}) - z_{12} z_{21}}$$

$$H_{\text{雙向網絡}} = \frac{v_L}{v_s} = \frac{R_L z_{21}}{(z_{11} + R_s)(R_L + z_{22})}$$

因此 Z 參量表述的條件 = 端口網絡單向化條件是 $|z_{12} z_{21}| \ll |(z_{11} + R_s)(R_L + z_{22})|$



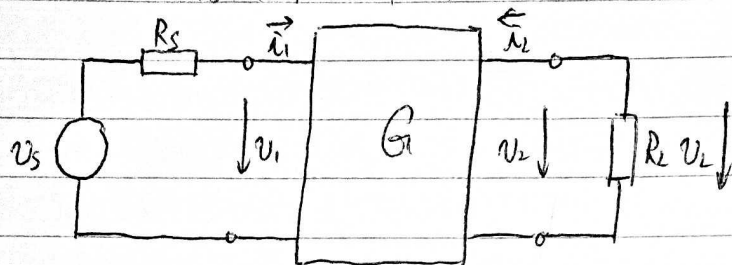
$$\begin{pmatrix} v_s - i_1 R_s \\ -\frac{v_L}{R_L} \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ v_L \end{pmatrix}$$

$$-\frac{v_L}{R_L} = i_1 h_{21} + v_L h_{22} \Leftrightarrow i_1 = -\frac{\frac{1}{R_L} + h_{22}}{h_{21}} v_L$$

$$v_s - i_1 R_s = i_1 h_{11} + v_L h_{12} \Leftrightarrow v_s = \left[-\frac{\frac{1}{R_L} + h_{22}}{h_{21}} (R_s + h_{11}) + h_{12} \right] v_L$$

$$\Rightarrow H_{\text{双向网络}} = \frac{V_L}{V_S} = \frac{h_{21}}{h_{12}h_{21} - (R_S + h_{11})(G_L + h_{22})}$$

同理可得, 当 $|h_{12}h_{21}| \ll |(R_S + h_{11})(G_L + h_{22})|$ 时, h 参量表示的线性二端口可视为单向网络



$$\begin{pmatrix} I_1 \\ V_L \end{pmatrix} = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \begin{pmatrix} V_S - I_1 R_S \\ -\frac{V_L}{R_L} \end{pmatrix}$$

$$V_L = V_S g_{11} - I_1 R_S g_{21} - \frac{V_L}{R_L} g_{22} \Leftrightarrow I_1 = \frac{V_S g_{11} - V_L (G_L g_{22} + 1)}{R_S g_{21}}$$

$$I_1 = V_S g_{11} - I_1 R_S g_{21} - \frac{V_L}{R_L} g_{22} \Leftrightarrow V_S g_{11} - (1 + R_S g_{21}) \cdot \frac{V_S g_{11} - V_L (G_L g_{22} + 1)}{R_S g_{21}} - \frac{V_L}{R_L} g_{22} = 0$$

$$\Leftrightarrow V_S \left(g_{11} - (G_S + g_{11}) \right) - \left(\frac{G_L G_S g_{22}}{g_{21}} + g_{12} G_L \right) V_L = 0$$

$$\Leftrightarrow V_S (g_{11} - (G_S + g_{11})) + \left[(1 + R_S g_{11}) \cdot \frac{G_L g_{22} + 1}{R_S g_{21}} - g_{12} G_L \right] V_L = 0$$

$$\Leftrightarrow V_S G_S + \left[\left(\frac{G_S + g_{11}}{g_{21}} \right) g_{22} + g_{12} \right] G_L V_L = 0$$

$$\Leftrightarrow \frac{V_L}{V_S} = \frac{G_S}{\left[(G_S + g_{11}) \frac{g_{22}}{g_{21}} + g_{12} \right] G_L}$$

$$\Leftrightarrow V_S G_S + \left[(G_S + g_{11}) \frac{G_L g_{22} + 1}{g_{21}} - g_{12} G_L \right] V_L = 0$$

$$\Leftrightarrow V_S G_S g_{21} = \left[(G_S + g_{11})(G_L g_{22} + 1) - g_{12} G_L g_{21} \right] V_L$$

$$\Leftrightarrow \frac{V_L}{V_S} = \frac{G_S g_{21}}{(G_S + g_{11})(G_L g_{22} + 1) - g_{12} g_{21} G_L} = \frac{G_S g_{21}}{R_L G_S g_{21} + (G_S + g_{11})(g_{22} + R_L) - g_{12} g_{21}}$$

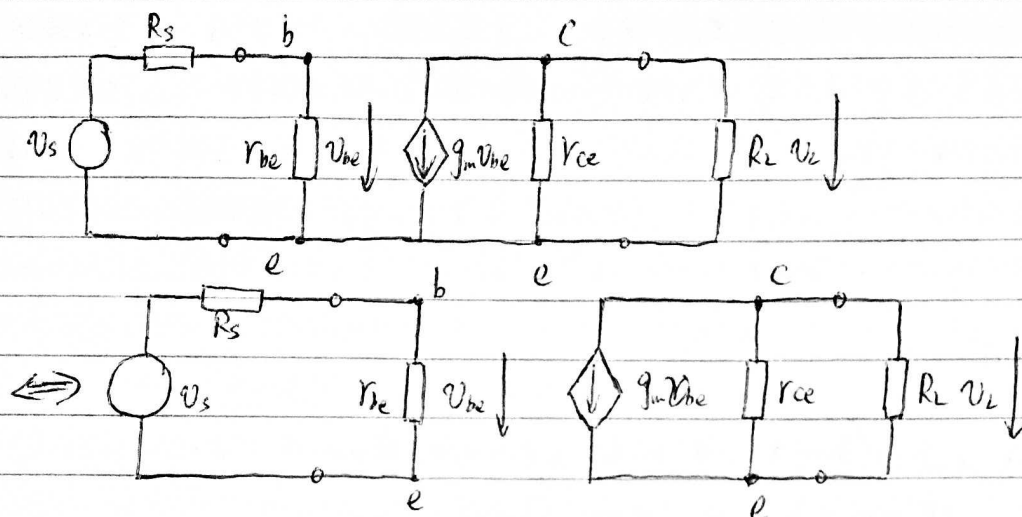
所以, $|g_{12}g_{21}| \ll |(G_S + g_{11})(g_{22} + R_L)|$ 是 g 参量表示的线性二端口网络单向化条件

年級：

科目：

作業 4.

(i) Common Emitter :



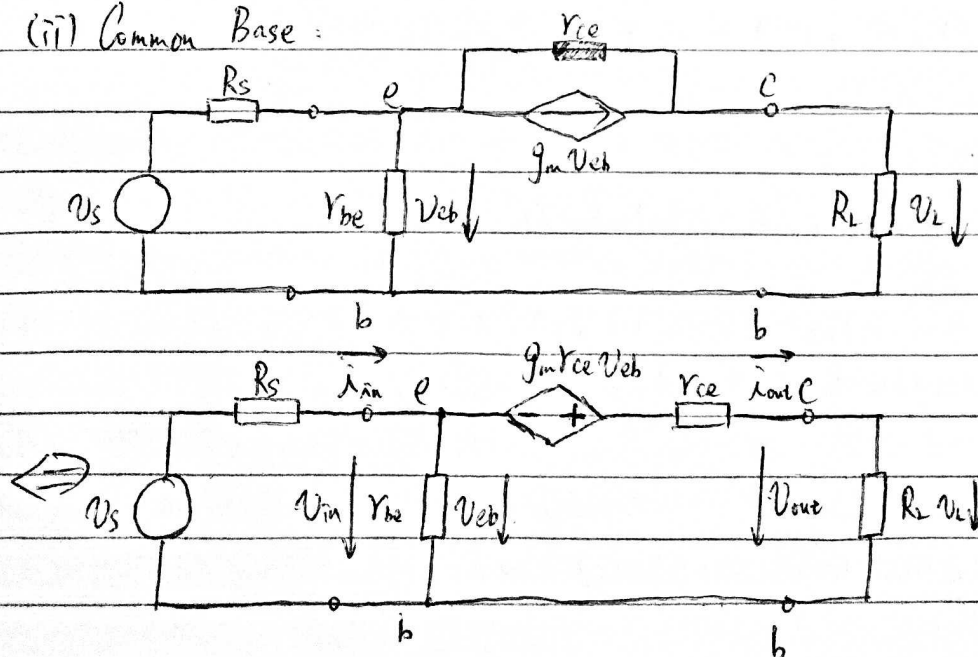
$$Z_{in} = r_{be}, \quad Z_{out} = r_{ce}$$

$$V_L = g_m r_{ce} V_{be} \cdot \frac{R_L}{r_{ce} + R_L} = \frac{r_{ce} R_L}{r_{ce} + R_L} g_m \cdot \frac{r_{be}}{R_s + r_{be}} V_s$$

因此.

$$Z_{in} = 10k\Omega, \quad Z_{out} = 100k\Omega, \quad \text{電壓放大倍數 } \frac{V_L}{V_s} = \frac{100 \times 10}{100 + 10} \times 10^{-3} \times 40 \times 10^{-3} \times \frac{10 \times 10^3}{50 + 10 \times 10^3} = 39.407$$

(ii) Common Base :

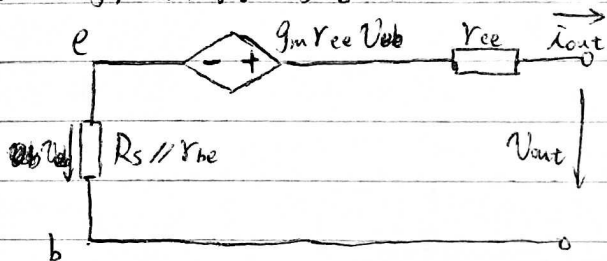


$$-g_m r_{ce} v_{eb} + \left(\hat{i}_{in} - \frac{v_{eb}}{r_{be}} \right) (r_{ce} + R_L) = v_{eb}, \quad v_{eb} = v_{in}$$

$$\Leftrightarrow (Y_{in} - G_{be}) (r_{ce} + R_L) - g_m r_{ce} = 1$$

$$\Leftrightarrow Y_{in} = \frac{g_m r_{ce} + 1}{r_{ce} + R_L} + G_{be} \Leftrightarrow Z_{in} = \frac{r_{ce} + R_L}{r_{ce} (g_m + G_{be}) + R_L G_{be}} = 25.186 \Omega$$

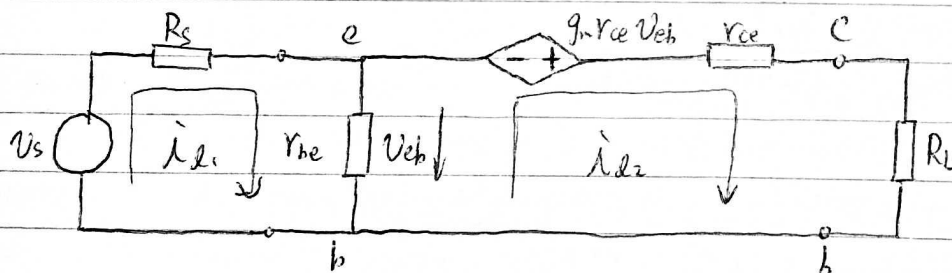
當 $U_s = 0$ 時，由輸出端看入電路時，電路等效於



$$i_{out} r_{ce} + v_{out} + i_{out} (R_s // r_{be}) - g_m r_{ce} (-i_{out} (R_s // r_{be})) = 0$$

$$v_{out} + i_{out} (r_{ce} + (R_s // r_{be}) + g_m r_{ce} (R_s // r_{be})) = 0$$

$$Z_{out} = -\frac{v_{out}}{i_{out}} = r_{ce} + (1 + g_m r_{ce}) \cdot \frac{R_s r_{be}}{R_s + r_{be}} = 297.95 k\Omega$$



$$U_s = \hat{i}_{L1} (R_s + r_{be}) - \hat{i}_{L2} r_{be}$$

$$0 = \hat{i}_{L2} (r_{ce} + R_L + r_{be}) - \hat{i}_{L1} r_{be} - g_m r_{ce} v_{eb}$$

$$v_{eb} = (\hat{i}_{L1} - \hat{i}_{L2}) r_{be}$$

$$\hat{i}_{L1} (R_s + r_{be}) - \hat{i}_{L2} r_{be} = U_s$$

$$\hat{i}_{L2} (r_{ce} + R_L + r_{be}) - \hat{i}_{L1} r_{be} - g_m r_{ce} r_{be} \hat{i}_{L1} + g_m r_{ce} r_{be} \hat{i}_{L2} = 0$$

$$\begin{pmatrix} R_s + r_{be} & -r_{be} \\ r_{be} (1 + g_m r_{ce}) & -(r_{ce} + r_{be} + R_L + g_m r_{ce} r_{be}) \end{pmatrix} \begin{pmatrix} \hat{i}_{L1} \\ \hat{i}_{L2} \end{pmatrix} = \begin{pmatrix} U_s \\ 0 \end{pmatrix}$$

$$\Rightarrow \hat{i}_{L2} = \frac{(-r_{be} (1 + g_m r_{ce}) \quad R_s + r_{be})}{r_{be}^2 (1 + g_m r_{ce}) - (R_s + r_{be}) (r_{ce} + r_{be} + R_L + g_m r_{ce} r_{be})} \begin{pmatrix} U_s \\ 0 \end{pmatrix}$$

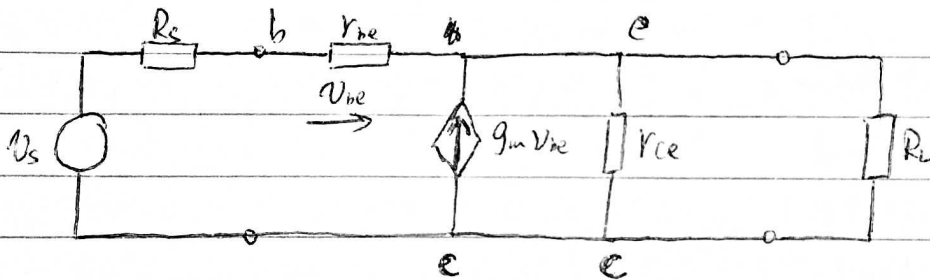
年級：

科目：

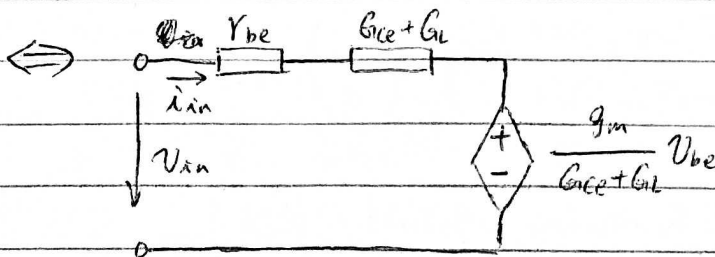
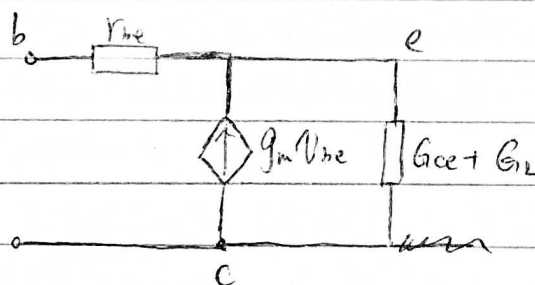
因此，電壓放大倍數

$$\begin{aligned}\frac{V_L}{V_S} &= \frac{\hat{i}_L R_L}{V_S} = \frac{R_L r_{be} (1 + g_m r_{ce})}{(R_S + r_{be})(r_{ce} + r_{be} + R_L + g_m r_{ce} r_{be}) - r_{be}^2 (1 + g_m r_{ce})} \\ &= \frac{R_L}{(R_S + r_{be}) \cdot \left(\frac{r_{ce} + R_L}{r_{be} (1 + g_m r_{ce})} + 1 \right) - r_{be}} = 13.268\end{aligned}$$

(iii) Common Collector



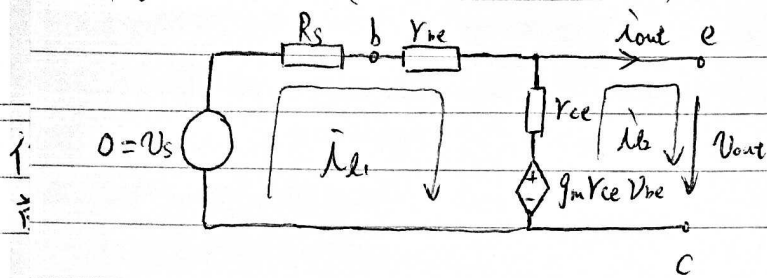
從輸入端看入



$$V_{in} = \hat{i}_{in} \left(r_{be} + \frac{1}{g_{ce} + g_L} \right) + \frac{g_m}{g_{ce} + g_L} r_{be} \hat{i}_{in}$$

$$Z_{in} = r_{be} + \frac{1 + g_m r_{be}}{g_{ce} + g_L} = 407.03 \Omega$$

從輸出端看入 (求 Z_{out} , 使得 $U_s = 0$)



$$\lambda_{i1} (R_s + r_{be} + r_{ce}) - \lambda_{i2} r_{ce} + g_m r_{ce} \lambda_{i1} r_{be} = U_s = 0$$

$$\lambda_{i2} r_{ce} + U_{out} - g_m r_{ce} \lambda_{i1} r_{be} - \lambda_{i1} r_{ce} = 0$$

$$\begin{pmatrix} R_s + r_{be} + r_{ce} + g_m r_{ce} r_{be} & -r_{ce} \\ r_{ce} (1 + g_m r_{be}) & -r_{ce} \end{pmatrix} \begin{pmatrix} \lambda_{i1} \\ \lambda_{i2} \end{pmatrix} = \begin{pmatrix} 0 \\ U_{out} \end{pmatrix}$$

$$\Rightarrow \lambda_{out} = \lambda_{i2} = \frac{\cancel{R_s + r_{be} + r_{ce} + g_m r_{ce} r_{be}}}{-r_{ce} (R_s + r_{be})} (-r_{ce} (1 + g_m r_{be}) \quad R_s + r_{be} + r_{ce} (1 + g_m r_{be})) \begin{pmatrix} 0 \\ U_{out} \end{pmatrix}$$

培道中學校

姓名： (號)

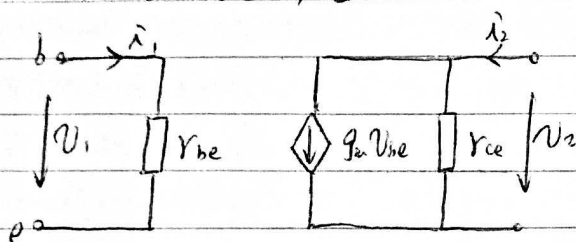
年級：

科目：

成績

作業 5

首先計算 Common Emitter 的 2 參量



$$Z_{11} = \left. \frac{V_1}{i_1} \right|_{i_2=0} = r_{be}$$

$$Z_{21} = \left. \frac{V_2}{i_1} \right|_{i_2=0} = \frac{-g_m V_{be} r_{ce}}{i_1} = -g_m r_{be} r_{ce}$$

$$Z_{12} = \left. \frac{V_1}{i_2} \right|_{i_1=0} = 0$$

$$Y = Z^{-1} = \frac{1}{(R_E + r_{be})(R_E + r_{ce}) - R_E(R_E - g_m r_{be} r_{ce})} \begin{pmatrix} R_E + r_{ce} & -R_E \\ -R_E + g_m r_{be} r_{ce} & R_E + r_{be} \end{pmatrix} = \begin{pmatrix} 0.07 & 0 \\ 2.83 & 0.007 \end{pmatrix} \text{ k}\Omega$$

$$\begin{aligned} Z_{in} = \frac{1}{y_{11}} &= \frac{R_E(r_{be} + r_{ce}) + r_{be}r_{ce} + R_E g_m r_{be} r_{ce}}{R_E + r_{ce}} = r_{be} + \frac{R_E r_{ce} (1 + g_m r_{be})}{R_E + r_{ce}} \approx r_{be} + \frac{R_E r_{ce} g_m r_{be}}{r_{ce}} \\ &= r_{be} + R_E g_m r_{be} \\ &= r_{be} (1 + g_m R_E) \\ &\approx r_{be} g_m R_E \end{aligned}$$

$$Z_{out} = \frac{1}{y_{22}} = \frac{R_E(r_{be} + r_{ce}) + r_{be}r_{ce} + R_E g_m r_{be} r_{ce}}{R_E + r_{be}} \approx r_{ce} g_m R_E$$

$$y_{21} = \frac{g_m r_{be} r_{ce} - R_E}{R_E(r_{be} + r_{ce}) + r_{be}r_{ce} (1 + g_m R_E)} \approx \frac{g_m r_{be} r_{ce}}{R_E(r_{be} r_{ce} + r_{be} + r_{ce})}$$

這樣具有負反饋的放大器的輸入電阻 Z_{in} 能夠趨向無窮，更接近理想放大器。